OPERATION PERFORMED: Right shoulder diagnostic arthroscopy, subacromial decompression with anterior acromioplasty, arthroscopic partial subscapularis debridement, and double-row transosseous-equivalent arthroscopic rotator cuff repair of the supraspinatus tendon.

INDICATION: A 54-year-old carpenter with persistent right shoulder pain, positive impingement tests, positive drop-arm sign, and MRI confirmation of a full-thickness 2.5 cm supraspinatus tear.

OPERATIVE DETAILS: Under regional interscalene nerve block supplemented by light general anesthesia, the patient was placed in the beach-chair position with the right arm supported in a Spider limb positioner. Standard posterior and anterior arthroscopic portals were established into the glenohumeral joint. Diagnostic arthroscopy revealed an intact glenoid labrum with mild superior fraying. The biceps tendon exhibited moderate tenosynovitis and was debrided. The humeral head articular cartilage was well-preserved. A complete U-shaped full-thickness tear of the supraspinatus tendon retracting to the glenoid rim was visualized.

The arthroscope was redirected into the subacromial space. Anterolateral and lateral working portals were placed under direct needle localization. A motorized shaver and radiofrequency ablator were used to perform extensive bursectomy. A hooked type-III acromion was reduced to a flat type-I contour utilizing an oval motorized burr, resecting approximately 5 mm of anterior-inferior acromial bone and releasing the coracoacromial ligament.

Attention was turned to the rotator cuff footprint. The greater tuberosity was decorticated to a bleeding cancellous bed with a motorized burr. The supraspinatus margin was mobilized by performing anterior and posterior interval slides. A double-row transosseous-equivalent construct was prepared. Two 4.5 mm PEEK medial row anchors preloaded with non-absorbable high-strength suture tape were placed at the articular cartilage margin of the greater tuberosity. Sutures were passed through the tendon substance in an inverted mattress fashion using a Viper suture passer. Two knotless lateral-row SwiveLock anchors were placed into the lateral cortical wall of the tuberosity, securing the crossed suture tapes under tension to generate a broad, flat footprint compression. Full glenohumeral range of motion verified tension-free stability of the repair. Portals were closed with 3-0 nylon sutures and the patient placed in an ultrasling with abduction pillow.